

Brain Tumor Segmentation

Mentor Review — Speaking Notes

What to say, slide by slide · keep it in your own words

0. Open with this (30 seconds)

Good morning sir. Our project is automated brain tumor segmentation on the BraTS2020 MRI dataset using a 3D U-Net in PyTorch. Beyond just training a model, we run a controlled ablation study — we compare four intensity-normalization methods and several augmentation strategies, including our own idea called tumor-aware CutMix. The full pipeline is built and verified; we are now moving to the training phase on Kaggle's GPUs.

1. Slide-by-slide script

Slide 1 — Title

Good morning sir. Our project is Brain Tumor Segmentation using a 3D U-Net on the BraTS2020 dataset. I am [name], presenting with my teammates.

Slide 2 — Table of Content

I will cover the motivation and problem, our objectives, existing work, our approach and implementation, how we evaluate it, and finish with the justification and conclusion.

Slide 3 — Introduction

Brain tumors are dangerous if not caught early. MRI is the main tool, but a radiologist outlining the tumor by hand is slow and can vary from person to person. Deep learning — specifically U-Net — can do this outlining automatically. Our motivation is an accurate, reproducible system, and to test which design choices actually make it better.

Slide 4 — Problem Statement

The problem is hard because tumors vary in size, shape, and location. Two extra challenges: MRI brightness is not standardized across machines, so preprocessing matters a lot; and the tumor is under 5% of the scan, so a model can look 95% accurate while completely missing it. Most projects just pick one recipe — we wanted to actually compare options.

Slide 5 — Objectives

Our objectives: build a 3D U-Net segmentation system; compare four normalization methods; test augmentation strategies including our own idea; evaluate with proper metrics — Dice, sensitivity, specificity, and HD95, not accuracy; and keep it reproducible with a leak-free patient-level split.

Slide 6 — Literature Review

U-Net and 3D U-Net are the standard architectures for medical segmentation. V-Net introduced Dice loss for imbalanced data. nnU-Net showed that preprocessing and augmentation choices often matter more than the architecture — which inspired our ablation study. CutMix is an augmentation we build on. BraTS is the benchmark everyone uses.

Slide 7 — Proposed System (text)

Our system takes the four MRI types as input, normalizes them, augments them, and feeds them to a 3D U-Net that outputs the tumor mask. We train with a combined Dice + Cross-Entropy loss to handle the imbalance. Our two contributions are the normalization comparison and our own augmentation, tumor-aware CutMix, which we are currently still developing.

Slide 8 — Proposed System (diagram) [point at each box]

This is the flow: the four MRI scans come in, we normalize the brightness, apply augmentation including our tumor-aware CutMix, then the 3D U-Net segments it. Because a full scan is too big for the GPU, we process it in patches and stitch them back together, then score the result with Dice and HD95.

Slide 9 — Implementation & Model Design

We load all 369 patients, normalize each scan over the brain region only, and split by patient so there is no data leakage. The model is a 3D U-Net with instance normalization, which works better than batch norm when the batch size is very small. We train in patches with efficiency techniques like mixed precision and early stopping, and at test time we slide a window across the full scan.

Slide 10 — Model Validation & Evaluation

To evaluate, we test on patients the model has never seen and compare its output to the doctor's ground-truth mask. We measure Dice for overlap, sensitivity and specificity for misses and false alarms, and HD95 for boundary accuracy — for all three tumor regions. We avoid accuracy because of the imbalance. These target numbers are from published results; our own numbers will come from our training runs, which are the next phase.

Slide 11 — Justification

This table summarizes why the approach is justified — accurate, automated, supports early detection, efficient, and importantly reproducible and rigorously validated with proper metrics rather than just accuracy.

Slide 12 — Conclusion

To conclude: we have built and verified a complete 3D U-Net pipeline with a controlled ablation study and our own tumor-aware CutMix augmentation, which we are currently finishing. Everything is reproducible. Our next step is running the full experiments on Kaggle's GPUs and reporting the real numbers.

Slide 13 — References [do not read out]

These are our main references — the BraTS benchmark papers, the U-Net and 3D U-Net architectures, V-Net for Dice loss, nnU-Net, and CutMix.

2. Your contribution — tumor-aware CutMix (say this well)

This is the part a mentor will dig into. Say it slowly, in your own words.

Standard CutMix mixes two scans by pasting a random patch, but since most of a brain scan is healthy tissue, that patch usually misses the tumor entirely. Our version pastes the patch directly on the tumor region, so every augmented example actually contains tumor — which should help with the class imbalance. We have implemented the first version and are now testing and refining it before the full experiments.

If asked: "isn't every sample having a tumor also an imbalance?"

The imbalance is at the voxel level within each scan, not across patients — and in BraTS every patient already has a tumor. Our CutMix pastes a small tumor region into a new scan, so the volume stays mostly background and realistic. We increase the variety of tumor the model sees, not the balance — we actually want it background-dominated so the model doesn't over-predict tumor on real scans.

3. Honest status — results (very important)

- Do NOT present any Dice or accuracy number as achieved. We have not done real training yet.
- We only ran a tiny end-to-end test (4 patients, 2 epochs, on CPU) to check the code works. Its scores are near zero, which is expected — it was just a bug-check, not a real run.

If asked "have you trained it / what are your results?", say:

We have run a small end-to-end test to validate the pipeline — it confirmed the model trains, the loss decreases, inference reconstructs the full scan, and it computes proper Dice and HD95 scores. The actual accuracy is near zero because the run was tiny and just for testing. Real training with meaningful results is our next step on Kaggle's GPU.

One line to remember: "The pipeline is validated; the numbers come next."

4. Likely questions & crisp answers

Q: Why is this more than a standard project?

The ablation study and our original tumor-aware CutMix — we test which choices help, not just train one model.

Q: Why not just use accuracy?

Severe class imbalance makes accuracy meaningless; Dice and HD95 are the real measures.

Q: How do you know results aren't luck?

Multiple random seeds, and we report mean plus/minus standard deviation.

Q: What is your baseline / comparison?

nnU-Net is the state-of-the-art reference; we are not claiming to beat it, we isolate the effect of preprocessing and augmentation choices.

Q: What is data leakage and how do you avoid it?

Neighbouring slices of one patient look almost identical; if they land in both train and test the score is faked. We split by whole patient, never mixing a patient across sets.

Q: Why instance normalization?

3D scans force a batch size of 1-2, where batch-norm statistics are unreliable; instance norm normalizes each scan on its own, so it works at any batch size.

Q: Have you trained it yet?

The pipeline is verified end-to-end; full training runs are our immediate next step.

Q: What are the limitations?

Single architecture, patch-based, non-official train/test split, and one dataset.

5. Questions to ask your mentor (shows engagement)

- Should we run the full 9-arm ablation, or a reduced 3-arm version given our compute budget?
- For the paper, is an arXiv preprint or a student conference the better target?
- Would you like us to add a comparison against a plain 2D U-Net baseline?

6. Delivery tips

- Slides 4, 7, and 8 are your key slides — slow down there.
- Speed through slides 2, 11, and 13 — they are supporting slides.
- Say the 'still developing' and 'results are next' lines confidently — they are normal for a mid-project review, not weaknesses.
- Close with: "That's our progress so far — happy to answer any questions."